

# Jacob Mattie

Vancouver  
778-710-7554  
j\_mattie@live.ca  
<https://github.com/qimbet>

## Summary

Data and software-oriented engineer with a foundation in Python, SQL, and Linux. Experienced in building ETL pipelines, automating workflows, and deploying scalable systems. Skilled at bridging software, hardware, and data architectures to deliver efficient, maintainable solutions.

**Software:** C++, Python, SQL, Redis, Selenium, Linux, Bash, Git

**Key Skills:** Nonlinear Optimization, Statistics, Linear Algebra

---

## Work Experience

### Investment Analyst

*Aligned Capital Partners / River Wealth*

Vancouver, BC

11/2024 - Ongoing

- Developed scheduled **web scraping and ETL pipelines** with Celery Beat and Redis, centralizing a custom PostgreSQL database from disparate public and proprietary sources.
- Built an end-to-end data pipeline to extract, clean, and load data from over 15 heterogeneous sources (**files, APIs, web scraping**) into a custom-designed SQL schema, laying the foundation for scalable data analytics.
- Configured and maintained a **Linux-based server stack** (Fedora, Gunicorn, Nginx) to platform a Django application, making a roster of data analytics, reporting, and automation tools accessible to non-technical staff.
- Automated recurring operational tasks in Python, eliminating 20+ hours of manual work per month and improving company-wide efficiency by 5%.

### Test Technologist

*IBC Technologies*

Burnaby, BC

6/2022 – 9/2023

- Automated hardware testing by merging IoT-integrated esp32 microcontrollers with PLC devices, opening **cloud-connected data pipelines** and enabling remote control of testing rigs for advanced lab automation.
- Automated analysis of high-volume datasets (40,000+ lines per series) using MATLAB, reducing processing time from 20 minutes to under 3 minutes per iteration, enabling rapid, reproducible **insight generation**.
- Engineered and deployed laser-based obstruction sensors and a responsive control system to upgrade an assembly line, boosting boiler lift speed by 25% while simultaneously enhancing operator safety.

### Opinions Editor

*The Peak Publications Society*

Burnaby, BC

10/2020 – 6/2022

- Managed the Opinions Section of a weekly newspaper, **coordinating a rotating team** of contributors and production staff to publish 2500 words across 5 articles while meeting strict production deadlines.
- **Mentored and coached** writers of varying experience levels, providing editorial feedback, thematic guidance, and structural training to develop their voice and align their work with the section's goals.

## IT Technician (Internship)

*BC Cancer Research Centre*

Vancouver, BC

09/2018 – 01/2019

- Diagnosed and resolved complex technical issues across **Windows, Mac, and Linux** environments in a large-scale research facility, showing adaptability and rapid problem-solving skills with diverse technology stacks.
  - Configured and maintained network infrastructure (switches, routers, cables) to ensure **cybersecurity** compliance and reliable data communication across research departments.
  - Created internal IT guides covering daily operations and network/domain structure, closing a key knowledge gap and **standardizing support processes**.
- 

## Projects

### Hydroponics

*Independent Project*

04/2025 – 08/2025; Ongoing

- Designed and assembled a modular hydroponics system able to grow 36 spinach plants from seed to harvest with no manual input.
  - Engineered esp32-based circuitry with relays, sensors, and power converters to control grow lights, watering pumps, and fertilizer dispersion systems.
  - Planning integration of a robotic arm to automate strawberry pollination using computer vision techniques.
- 

## Education

### Simon Fraser University

*B.A.Sc in Mathematics*

01/2019 - 05/2022

---

## Certifications

### Nuclear Energy Worker

*Canadian Nuclear Safety Commission*

09/2025 - 09/2028

### AI and Technology Foundations

*Canada's Ocean Supercluster*

04/2025 - 06/2025

---

## Volunteering & Leadership

### Triumph - ASPIRE Astrochemistry Lab

*Automation Technologist*

Vancouver, British Columbia

05/2025 - Ongoing

- Developed a **Bash-based installer** for EPICS, a Flask-backed control framework widely used in physics laboratories for managing **data acquisition** and actuator systems.
- Integrated a Python motor controller into the laboratory control system, **reducing experimental variance to 0%** and eliminating need for active engagement with the experiment over its hours-long runtime.